

British Airways

Task 1

What British Airways wants

British Airways provides special waiting areas (called **lounges**) for certain passengers at **Terminal 3 of Heathrow Airport** in London.

A lounge is more comfortable than the normal airport waiting area. Passengers can:

- Sit in comfortable seats
- Eat free food and drinks
- Use Wi-Fi
- Work or relax before their flight
- Sometimes take a shower or use business facilities

For example, if a passenger is flying Business Class, First Class, or has a high loyalty status with British Airways, they may be allowed to enter the lounge before boarding their flight.

So, Not every passenger can use them. Different passengers get access based on:

- Their ticket class (Business, First Class, etc.)
- Their loyalty status (Gold, Silver, etc.)
- Other eligibility rules

BA wants to **predict how many passengers will be eligible for each lounge type in the future.**

The problem

British Airways does **not know exactly what future flights will look like** because:

- Flight schedules can change.
- New routes may be added.
- Some routes may be removed.
- Different aircraft may be used.

So they cannot say:

On the flight named **BA123**, about **25 passengers** are allowed to use the **British Airways lounge** before their flight.

because that specific flight might not exist in the future.

Your job

Think of this task as if **British Airways** is asking:

"We don't know exactly what future flights will be. Can you create a simple guide that tells us approximately how many passengers on different types of flights will be able to use our lounges?"

Step 1: Group similar flights

Instead of looking at every flight individually, put flights into categories.

Flight Group	Example
Europe	London → Paris
North America	London → New York
Asia	London → Singapore
Middle East	London → Dubai
Africa	London → Johannesburg

Why? Because passengers on flights to the same region often have similar travel patterns.

Step 2: Estimate lounge eligibility

For each group, estimate:

- How many passengers might use the Concorde Room
 - How many might use the First Lounge
 - How many might use Club Lounge
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Step 3: Download the flight schedule

You will get a spreadsheet containing flights departing from Heathrow Terminal 3.

What the columns mean

Column Name	Meaning
FLIGHT_DATE	Date of the flight
FLIGHT_TIME	Departure time of the flight
TIME_OF_DAY	Morning, Afternoon, Evening, etc.
AIRLINE_CD	Airline code (BA = British Airways)
FLIGHT_NO	Flight number
DEPARTURE_STATION_CD	Airport the flight leaves from (LHR = Heathrow)
ARRIVAL_STATION_CD	Airport the flight arrives at (JFK, LAX, FRA, etc.)
ARRIVAL_COUNTRY	Destination country
ARRIVAL_REGION	Destination region (Europe, Asia, North America, etc.)
HAUL	Flight type: SHORT or LONG haul
AIRCRAFT_TYPE	Aircraft used (A320, B777, A380, etc.)
FIRST_CLASS_SEATS	Number of First Class seats
BUSINESS_CLASS_SEATS	Number of Business Class seats
ECONOMY_SEATS	Number of Economy seats
TIER1_ELIGIBLE_PAX	Passengers eligible for Tier 1 (Concorde Room)
TIER2_ELIGIBLE_PAX	Passengers eligible for Tier 2 (First Lounge)
TIER3_ELIGIBLE_PAX	Passengers eligible for Tier 3 (Club Lounge)

What you should do in Excel

Step 1: Open the Excel file

Step 2: Create a new column called "TOTAL_SEATS"

Insert a column between existing columns

If you want the column after ECONOMY_SEATS:

1. Right-click the column header TIER1_ELIGIBLE_PAX.
2. Click Insert.
3. A new blank column appears before TIER1_ELIGIBLE_PAX.
4. Name it:

TOTAL_SEATS

Add the formula

Suppose:

- FIRST_CLASS_SEATS is column L
- BUSINESS_CLASS_SEATS is column M
- ECONOMY_SEATS is column N
- TOTAL_SEATS is column O

In the first row under TOTAL_SEATS enter:

=L2+M2+N2

Copy the formula to all rows

1. Click the cell containing the formula.
2. Move your mouse to the small square in the bottom-right corner.
3. Double-click it.

Excel automatically fills the formula down for all flights.

Next columns

After TOTAL_SEATS, create:

Column Name
TIER1_%
TIER2_%
TIER3_%

Formula for TOTAL_SEATS (if needed)

If:

- L = FIRST_CLASS_SEATS
- M = BUSINESS_CLASS_SEATS
- N = ECONOMY_SEATS

Then in O2:

=L2+M2+N2

Columns acc to excel file:

Column	Name
O	TOTAL_SEATS
P	TIER1_ %
Q	TIER2_ %
R	TIER3_ %
S	TIER1_ELIGIBLE_PAX
T	TIER2_ELIGIBLE_PAX
U	TIER3_ELIGIBLE_PAX

Use these formulas:

TIER1_ % (Cell P2)

=S2/O2

TIER2_ % (Cell Q2)

=T2/O2

TIER3_ % (Cell R2)

=U2/O2

After entering the formulas

1. Press Enter
2. Select columns P, Q, R

3. Go to Home → Percentage (%)

4. Increase decimals to 2 places

Example output:

TIER1_%	TIER2_%	TIER3_%
0.00%	4.26%	16.17%
0.30%	3.63%	12.69%
0.56%	6.11%	21.67%

The reason for these formulas is very simple.

British Airways wants to know:

What percentage of passengers on a flight are eligible for each lounge tier?

Tier 1 Formula

Formula

=S2/O2

Where:

- S2 = TIER1_ELIGIBLE_PAX
- O2 = TOTAL_SEATS

Why?

Because:

Tier 1 Eligible Passengers

Total Passengers

gives the percentage of passengers eligible for Tier 1.

Meaning:

0.43% of passengers on this flight qualify for Tier 1 (Concorde Room).

Tier 2 Formula

Formula

$$=T2/O2$$

Where:

- $T2 = \text{TIER2_ELIGIBLE_PAX}$
- $O2 = \text{TOTAL_SEATS}$

Why?

Because:

Tier 2 Eligible Passengers

Total Passengers

gives the percentage eligible for the First Lounge.

Meaning:

About 4.26% of passengers can use the First Lounge.

Tier 3 Formula

Formula

$$=U2/O2$$

Where:

- $U2 = \text{TIER3_ELIGIBLE_PAX}$
- $O2 = \text{TOTAL_SEATS}$

Why?

Because:

Tier 3 Eligible Passengers

Total Passengers

gives the percentage eligible for the Club Lounge.

Meaning:

About 16.17% of passengers can use the Club Lounge.

Why BA wants percentages instead of passenger counts

Passenger counts change from flight to flight.

For example:

Flight	Seats	Tier3 Eligible
Flight A	235	38
Flight B	469	75

The raw numbers are different.

This percentage can then be applied to future flights even if:

- Flight numbers change
- Aircraft types change
- Seat capacity changes

That's why you're calculating:

$\text{Eligible Passengers} \div \text{Total Seats} \times 100$

for Tier 1, Tier 2, and Tier 3. This converts passenger counts into reusable percentages, which is exactly what the lookup table needs.

Now do the next step: Create a Pivot Table

Step 1

Select your entire dataset.

Shortcut:

Ctrl + A

Step 2

Click:

Insert → PivotTable

Choose:

New Worksheet

Click OK

Step 3

In the PivotTable Fields panel:

Rows

Drag:

ARRIVAL_REGION

into Rows

Step 4

In Values drag:

TIER1_%

Under Values, you have:

Sum of TIER1_%

To change it to Average

- 1. In the Values box (bottom-right), find:**

Sum of TIER1_%

- 2. Click the small ▼ arrow next to Sum of TIER1_%.**

- 3. Select:**

Value Field Settings...

(If you don't see it immediately, right-click on one of the numbers in the Pivot Table, such as 2053.8889%, and choose Value Field Settings.)

- 4. A popup window opens.**
- 5. Under Summarize Values By, choose:**

Average

- 6. Click OK.**
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Why?

Your current numbers are huge:

Region Sum of TIER1_ %

Europe 2053.89%

This is adding all flight percentages together, which doesn't make sense.

You want:

Region Average Tier1 %

Europe ~0.5%–1%

because BA wants the typical percentage per flight, not the total.

After that

Repeat the same process for:

- **TIER2_ %**
- **TIER3_ %**

So your final Pivot Table will show:

ARRIVAL_REGION	Average Tier1 %	Average Tier2 %	Average Tier3 %
Asia	...	...	...
Europe	...	...	...
Middle East	...	...	...
North America	...		

Region	Average of TIER1 %	Average of TIER2 %	Average of TIER3 %
Asia	0.3965%	2.6667%	10.9799%
Europe	0.3437%	4.3099%	16.9000%
Middle East	0.2571%	2.6596%	10.8068%
North America	0.2083%	2.5489%	11.2000%
Grand Total	0.2829%	3.6629%	14.5483%

Justification: Flights were grouped by region (Europe, North America, Asia, and Middle East) because passengers traveling to different regions may have different lounge usage patterns. The percentage of eligible passengers was calculated for each lounge tier and

Justification

Prompt	Your Response
1. How did you group the flights in your table?	I grouped flights by destination region: Europe, North America, Asia, and Middle East.
2. Why did you choose that grouping method?	Different regions may have different types of passengers and lounge demand. This method is simple and easy to use.
3. What assumptions did you make about passenger eligibility?	I assumed that the percentage of lounge-eligible passengers varies by region. The percentages were calculated using historical flight data.
4. How can your model apply to future or changing flight schedules?	The model uses regional averages, so it can be used even if flight numbers, routes, or aircraft change in the future.